lecture-15

$$\langle f, g\rangle = \int_{-\pi}^{\pi} fg\,dx$$

$$a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(kx)\,dx$$

$$b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(kx)\,dx$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k\cos(kx) + \sum_{k=1}^{\infty} b_k\sin(kx)$$

2π periodic function $(-\pi,\pi)$

In General Interval

$$\langle f, g\rangle = \int_{-L}^{L} f(x)g(x)\,dx$$

$$a_k = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(kx\frac{2\pi}{L}\right)dx$$

$$b_k = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(kx\frac{2\pi}{L}\right)dx$$

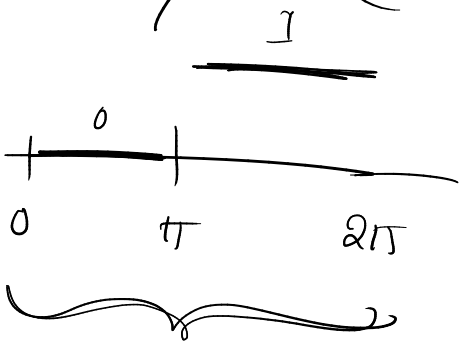
$$a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\,dx$$

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k\cos\left(k\frac{x\pi}{L}\right) + \sum_{k=1}^{\infty} b_k\sin\left(k\frac{x\pi}{L}\right)$$

$2L$ periodic $(-L, L)$

Example

$$f(x) = \begin{cases} 1, & \pi < x < 2\pi \\ 0, & 0 < x < \pi \end{cases}$$



$$a_k = \frac{1}{\pi}\int_{0}^{2\pi} f(x)\cos\left(kx\frac{x\pi}{\pi}\right)$$

Half of the interval

function Domain

(Here we always follow the domain of the function not the general formula)

$2L = 2\pi$

$L = \pi$

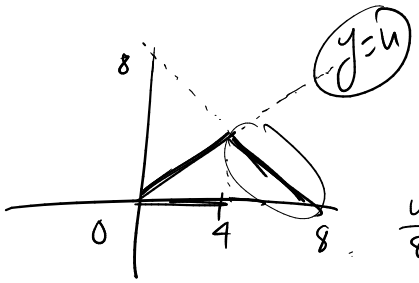
$$b_k = \frac{1}{\pi}\int_0^{2\pi} f(u)\sin\left(k\times\frac{2\pi}{\pi}\right)du$$

$$a_0 = \frac{1}{\pi}\int_0^{2\pi} f(u)\,du$$

$$f(u) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k\cos\left(k\times\frac{2\pi}{\pi}\right) + \sum_{k=1}^{\infty} b_k\sin\left(k\times\frac{2\pi}{L}\right)$$

Example

$$f(u) = \begin{cases} u, & 0<u<4 \\ 8-u, & 4<u<8 \end{cases}$$



$\frac{u}{8}+\frac{y}{8}=1$

$2L=(8-0)$

$L=4$

$y=8\left(1-\frac{u}{8}\right)$

$\boxed{y=8-u}$

$$a_k = \frac{1}{4}\int_0^8 f(u)\cos\left(k\times\frac{2\pi}{4}\right)du$$

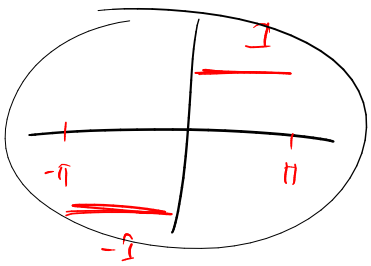
$$b_k = \frac{1}{4}\int_0^8 f(u)\sin\left(k\frac{u\pi}{4}\right)du$$

$$a_0 = \frac{1}{4}\int_0^8 f(u)\,du$$

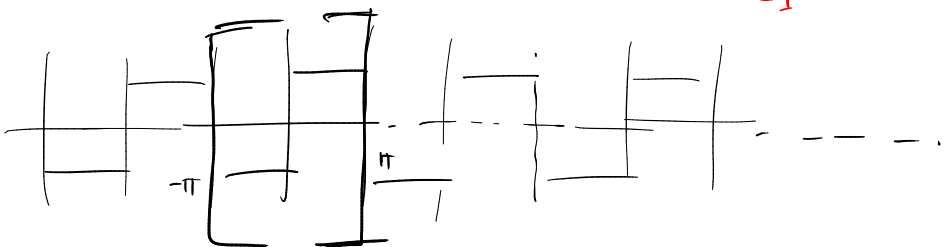
$$f(u) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k\cos\left(k\times\frac{2\pi}{4}\right) + \sum_{k=1}^{\infty} b_k\sin\left(\frac{ku\pi}{4}\right)$$

Recall:

$$f(u) = \begin{cases} 1, & 0<u<\pi \\ -1, & -\pi<u<0 \end{cases}$$



Periodic Extension



$$f(x) = \sum_{k=\text{odd}} \frac{4}{\pi k} \sin(kx)$$

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos(kx) + \sum_{k=1}^{\infty} b_k \sin(kx)$$

$$\left.\begin{aligned} e^{i\theta} &= \cos\theta + i\sin\theta \\ e^{-i\theta} &= \cos\theta - i\sin\theta \end{aligned}\right\} \quad \cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2} \;\&\; \sin\theta = \frac{e^{i\theta}-e^{-i\theta}}{2i}$$

Q/ Find the complex form of Fourier.

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \frac{e^{i(kx)} + e^{-i(kx)}}{2} + \sum_{k=1} b_k \frac{e^{i(kx)} - e^{-i(kx)}}{2i}$$

$$= \frac{a_0}{2} + \sum_{k=1}^{\infty} \frac{a_k}{2} e^{i(kx)} + \frac{a_k}{2} e^{-i(kx)} + \sum_{k=1}^{\infty} \frac{b_k}{2i} e^{i(kx)} - \frac{b_k}{2i} e^{-i(kx)}$$

$$= \frac{a_0}{2} + \sum_{k=1}^{\infty} \left(\frac{a_k}{2} + \frac{b_k}{2i}\right) e^{i(kx)} + \sum_{k=1}^{\infty} \left(\frac{a_k}{2} - \frac{b_k}{2i}\right) e^{-i(kx)}$$

$$= \frac{a_0}{2} 1 + \sum_{k=1}^{\infty} \frac{1}{2}(a_k - ib_k)\, e^{i(kx)} + \sum_{k=1}^{\infty} \frac{1}{2}(a_k + ib_k)\, e^{i(-k)x}$$

$$e^{i \times 0 \times x}$$

$$\left\{ e^{ikx} \right\}_{k=-\infty}^{\infty}$$

$$= \underbrace{\boxed{\frac{a_0}{2}}}_{c_0} + \sum_{k=1}^{\infty} \underbrace{\boxed{\frac{1}{2}(a_k - ib_k)}}_{c_k} e^{ikx} + \sum_{k=-1}^{-\infty} \underbrace{\boxed{\frac{1}{2}(a_{-k} + ib_{-k})}}_{c_{-k}} e^{ikx}$$

$k=0$ $\quad$ $k=1, \dots \infty$ $\quad$ $k=-1, \dots -\infty$

$$= \boxed{\sum_{k=-\infty}^{\infty} c_k e^{ikx}}$$

$$\text{Basis} = \left\{ 1, \cos(kx), \sin(kx) \right\}_{k=1}^{\infty}$$

$$\boxed{\sum_{k=-\infty}^{k=\infty} c_k e^{ikx}} \equiv f(x) \equiv \boxed{\frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos(kx) + \sum_{k=1}^{\infty} b_k \sin(kx)}$$

$$\left\{ e^{ikx} \right\}_{k=-\infty}^{\infty}$$

$$f(x) = \sum_{k=-\infty}^{\infty} c_k e^{ikx}$$

$$c_m = \frac{\langle f(x), e^{imx} \rangle}{\langle e^{imx}, e^{imx} \rangle} = \boxed{\frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\, e^{-imx}\, dx}$$

$$f(x) = \begin{cases} -1, & -\pi < x < 0 \\ 1, & 0 < x < \pi \end{cases}$$

2π – Periodic

$2L = 2\pi$

$L = \pi$

$$f(x) = \sum_{k=-\infty}^{\infty} c_k e^{ik\frac{x\pi}{L}}$$

$$f(x) = \sum_{k=-\infty}^{\infty} c_k e^{ikx}$$

$$c_k = \frac{1}{2L}\int_{-L}^{L} f(x) e^{-ik\frac{x\pi}{L}} dx$$

$$c_k = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x) e^{-ikx} dx$$

$$c_k = \frac{1}{2\pi}\left[\int_{-\pi}^{0} (-1) e^{-ikx} dx + \int_{0}^{\pi} (1) e^{-ikx} dx\right]$$

$$= \frac{1}{2\pi}\left(-\left[\frac{e^{-ikx}}{-ik}\right]_{-\pi}^{0} + \left[\frac{e^{-ikx}}{-ik}\right]_{0}^{\pi}\right)$$

$$= \frac{1}{2\pi}\left[-\left(\frac{1}{-ik} + \frac{e^{+ik\pi}}{ik}\right) + \left(\frac{e^{-ik\pi}}{-ik} + \frac{1}{ik}\right)\right]$$

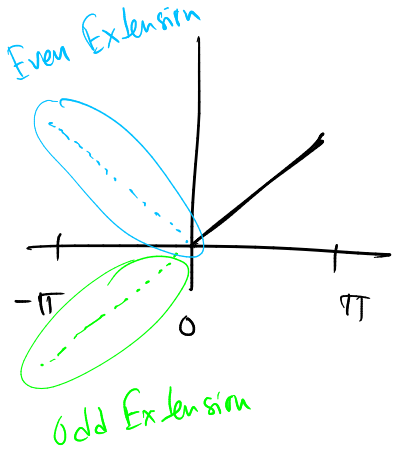
$$= \frac{1}{2\pi}\left(\frac{1}{ik} - \frac{e^{ik\pi}}{ik} - \frac{e^{-ik\pi}}{ik} + \frac{1}{ik}\right) = \frac{1}{2\pi}\left[\frac{2}{ik} - \frac{2(-1)^k}{ik}\right]$$

$$e^{ik\pi} = \cancel{\cos(k\pi)}^{(-1)^k} + i\,\cancel{\sin(k\pi)}^{0}$$

$$f(x) = \sum_{k=-\infty}^{\infty} \frac{1}{2\pi}\left(\frac{2}{ik} - \frac{2(-1)^k}{ik}\right) e^{ikx}$$

Half interval

$$f(x) = x, \quad 0 < x < \pi$$



Even function,

$$\hat{f}(x) = \begin{cases} x, & 0 < x < \pi \\ -x, & -\pi < x < 0 \end{cases}$$

Formula

$$\boxed{\hat{f}_{Even}(x) = \begin{cases} x & 0 < x < \pi \text{ given} \\ f(-x) & -\pi < x < 0, \text{ Extension} \end{cases}}$$

$$\hat{f}_{odd}(x) = \begin{cases} f(x), & 0 < x < \pi \quad \text{Given} \\ -f(-x), & -\pi < x < 0 \end{cases}$$

$$\hat{f}_{odd}(x) = \begin{cases} x, & 0 < x < \pi \\ x, & -\pi < x < 0 \end{cases}$$

$$-f(-x)$$

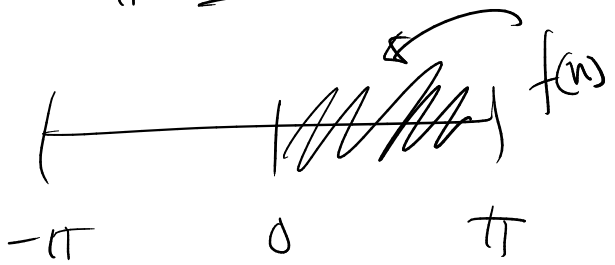
$$-(-x) = x$$

$f(x) \longrightarrow$ Fourier Series

$$\hat{f}_{odd}(x) \quad \boxed{\cancel{\frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos(kx)}} + \sum_{k=1}^{\infty} b_k \sin(kx)$$

$$\hat{f}_{even}(x) = \frac{a_0}{2} + \sum_{k=1} a_k \cos(kx) + \boxed{\cancel{\sum_{k=1}^{\infty} b_k \sin(kx)}}$$

$$a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} \hat{f}_{even}(x) \cos(kx)\,dx = \frac{1}{\pi} \times 2\int_{0}^{\pi} f(x)\cos(kx)\,dx$$



$$b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f_{even}(x) \sin(kx)\,dx = 0$$

$$a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f_{odd}(x) \cos(kx)\,dx = 0 \qquad a_0 = 0$$

$$b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f_{odd}(x) \sin(kx)\,dx = \frac{1}{\pi} \times 2\int_{0}^{\pi} f(x)\sin(kx)\,dx$$